K.Shamela

Mini-Project -1

Project Title : Phonepe Transaction Insights

Course : Data science

Batch : DS-C-WD-E-B144

PhonePe Transaction Insights

Problem Statement:

With the increasing reliance on digital payment systems like PhonePe, understanding the dynamics of transactions, user engagement, and insurance-related data is crucial for improving services and targeting users effectively. This project aims to analyze and visualize aggregated values of payment categories, create maps for total values at state and district levels, and identify top-performing states, districts, and pin codes.

Introduction

Digital payment platforms have transformed the financial ecosystem in India. Among them, PhonePe has emerged as one of the most widely used digital payment applications.

This project analyzes PhonePe transaction data to identify patterns, trends, and insights related to:

- Transaction growth
- User engagement
- Device usage
- Insurance adoption
- Regional transaction performance

The analysis helps understand how users interact with the platform and identify opportunities for business expansion.

Objectives of the Project:

The main objectives of this project are:

- To analyze transaction trends over time.
- To identify top performing states and districts.
- To study user engagement patterns.
- To analyze device brand dominance among users.
- To understand insurance adoption trends across regions.
- To identify growth opportunities for PhonePe.

Data Extraction:

Data extraction is the process of collecting and importing data from the source repository into the analysis environment for further processing and analysis.

For this project, the PhonePe Pulse dataset was extracted from the official PhonePe Pulse GitHub repository, which contains structured data related to digital payment transactions, user registrations, and insurance services across India.

The dataset is stored in a hierarchical folder structure containing JSON files organized by year, quarter, state, district, and pin code levels.

The following steps were performed to extract the data:

1. The PhonePe Pulse dataset was downloaded from the GitHub repository.
2. The dataset contained multiple JSON files organized in folders based on year and quarter.
3. Python libraries such as Pandas and JSON were used to read the JSON files.
4. The extracted data was converted into structured tabular format.
5. The cleaned data was then stored in a SQL database with 12 tables for efficient querying and analysis.

SQL Database and Table Creation:

Tables to store data from the different folders

Aggregated Tables

These tables contain **state-level summarized data** for transactions and users across different years and quarters.

- **Agg_transaction:** Contains aggregated transaction data including transaction type, transaction count, and transaction amount for each state and quarter.
- **Agg_user:** Stores aggregated user data including device brand and the number of registered users using different smartphone brands.
- **Agg_insurance:** Contains aggregated insurance transaction information including insurance transaction count and total insurance amount.

Map Tables

These tables provide **district-level geographical mapping data** for transactions, users, and insurance.

- **Map_transaction:** Holds transaction counts and transaction amounts mapped across districts.

- **Map_user:** Contains mapping information for registered users and number of app opens across districts.
- **Map_insurance:** Stores district-level insurance transaction count and amount.

Top Tables

These tables identify the **top performing regions** based on transaction activity, users, and insurance.

- **Top_transaction_district:** Contains the districts with the highest transaction counts and amounts.
- **Top_transaction_pincode:** Lists the pin codes with the highest transaction activity.
- **Top_user_district:** Contains districts with the highest number of registered users.
- **Top_user_pincode:** Lists pin codes with the highest registered user counts.

Insurance Tables

These tables highlight **top performing insurance regions**.

- **Top_insurance_district:** Contains districts with the highest insurance transaction counts and values.
- **Top_insurance_pincode:** Lists pin codes with the highest insurance transaction activity.

Tool Technologies Used:

Python for Data analysis, Pandas for Data cleaning and processing ,MySQL for Querying the Phonepe data, Matplotlib for data visualization, Seaborn for statistical visualization, Plotly for interactive charts and Streamlit for Dashboard creation.

Data Visualization:

Several visualization techniques were used to understand the data clearly:

- Line Charts → Trend analysis over time
- Bar Charts → Comparison between states or devices
- Pie / Donut Charts → Distribution analysis
- Horizontal Bar Charts → Ranking analysis

These charts help convert complex datasets into easy-to-understand insights.

Business Case Study:

1. Decoding Transaction Dynamics on PhonePe

Scenario

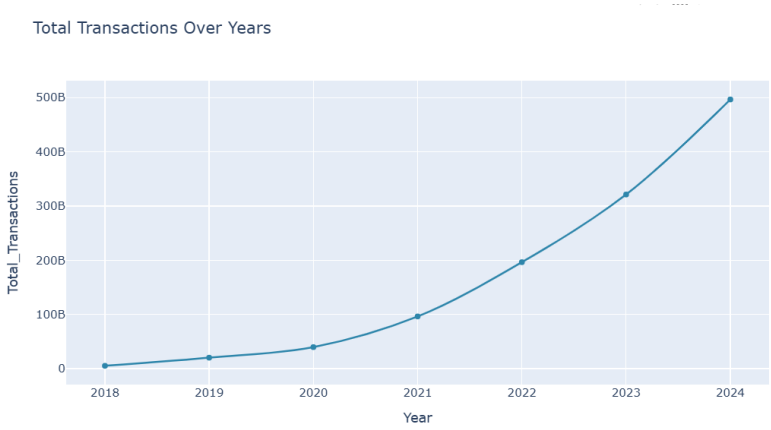
PhonePe, a leading digital payments platform, has recently identified significant variations in transaction behaviour across states, quarters, and payment categories. While some regions and transaction types demonstrate consistent growth, others show stagnation or decline. The leadership team seeks a deeper understanding of these patterns to drive targeted business strategies.

Objective: Understand transaction behaviour across years, states, and transaction types.

Insights:

Query:1. Total Transactions Over Years

- The number of PhonePe transactions increased steadily over the years.
- This indicates growing adoption of digital payments in India.
- The growth trend reflects increased user trust and smartphone penetration.



Key Insights:

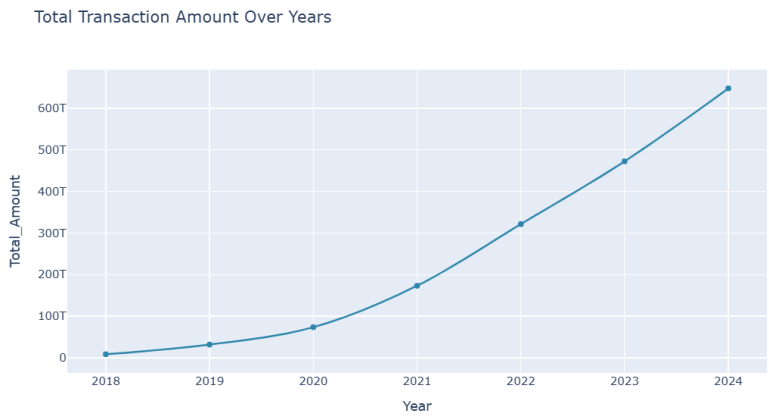
- Transactions have grown steadily, indicating increasing adoption of digital payments.
- Growth reflects higher user trust and smartphone penetration across India.

Recommendations:

- Focus on retaining existing users while expanding in emerging states.
- Introduce loyalty programs to encourage more frequent transactions.

Query:2. Total Transaction Amount Over Years

- Transaction value increased significantly each year.
- This shows that users are not only making more transactions but also transacting higher amounts.



Key Insights:

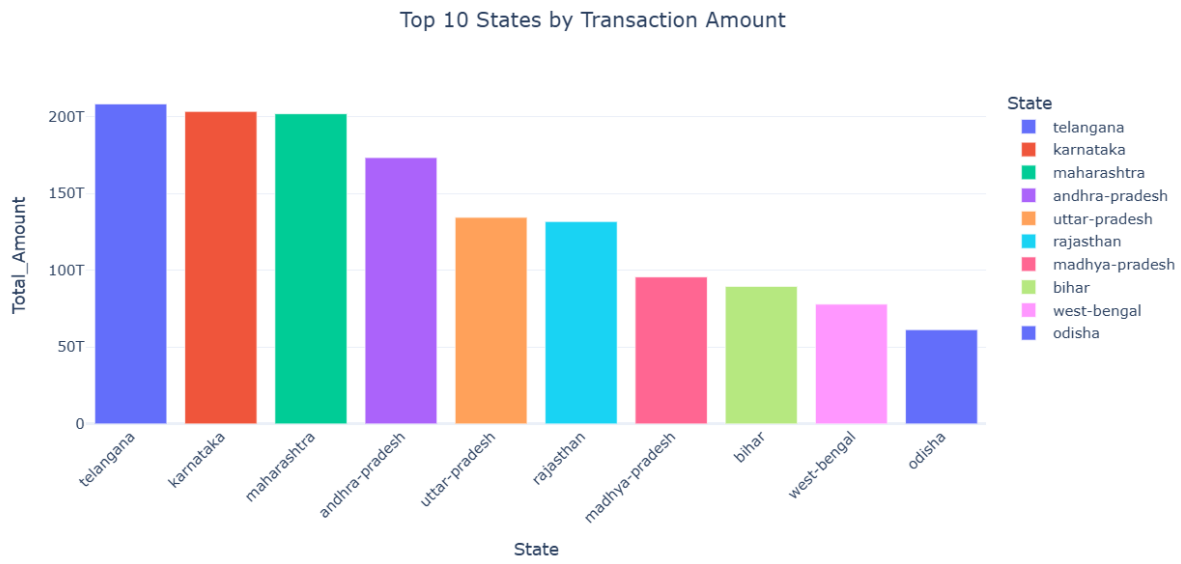
- Transaction values have increased, showing users are transacting larger amounts.
- Growth in transaction value complements the rise in transaction volume.

Recommendations:

- Promote higher-value transactions through targeted offers.
- Explore premium services for frequent high-value users.

Query:3. Top 10 States by Transaction Amount

- States like Maharashtra, Karnataka, and Telangana show higher transaction values.
- These states have strong digital infrastructure and higher smartphone usage



Key Insights:

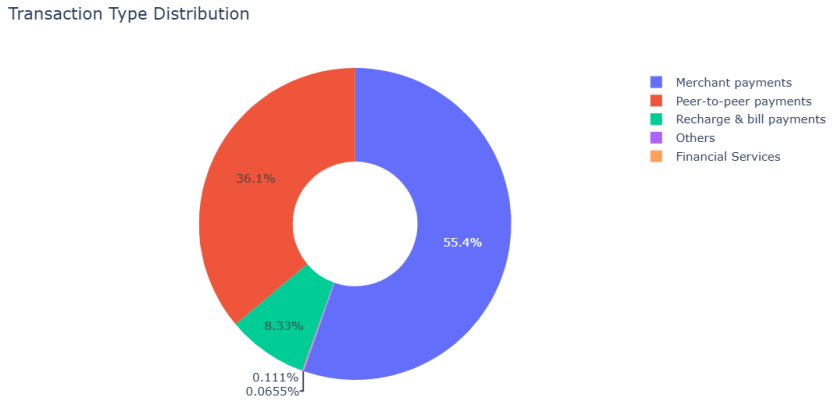
- Maharashtra, Karnataka, and Telangana lead in transaction amounts.
- Strong digital infrastructure and smartphone usage drive this trend.

Recommendations:

- Prioritize marketing and partnerships in top-performing states.
- Target low-performing states with localized campaigns to improve adoption and market penetration.

<u>Query:4.Transaction Type Distribution</u>

- Categories such as Peer-to-Peer (P2P) and Merchant Payments dominate the transaction types.
- This shows that PhonePe is widely used for daily payments and money transfers



Key Insights:

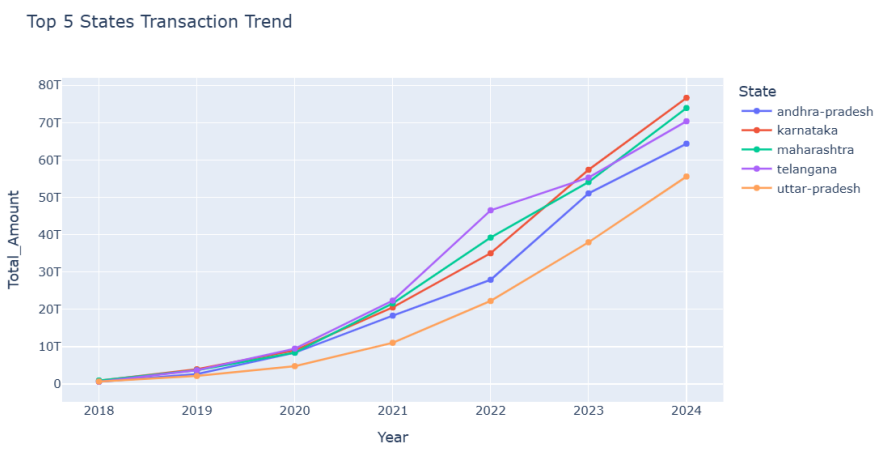
- Peer-to-Peer (P2P) and Merchant Payments dominate usage.
- PhonePe is a preferred platform for daily payments and money transfers.

Recommendations:

- Enhance user experience in top transaction categories.
- Introduce incentives for underutilized transaction types to diversify revenue.

Query:5. Top States Transaction Trend

- Some states consistently maintain high transaction activity.
- These states represent key markets for PhonePe expansion strategies.



Key Insight:
Transaction growth varies across states, highlighting regional differences in adoption.

Recommendation:
Focus on high-growth states for expansion and design targeted strategies for slow-growing regions.

2. Device Dominance and User Engagement Analysis

Scenario

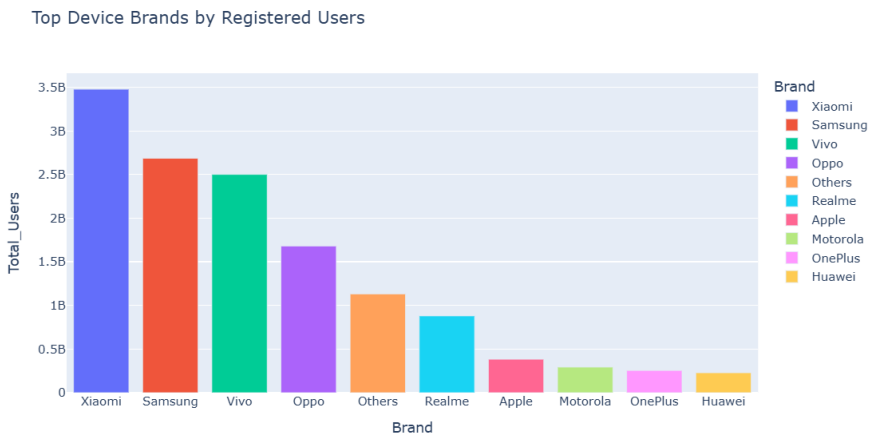
PhonePe aims to enhance user engagement and improve app performance by understanding user preferences across different device brands. The data reveals the number of registered users and app opens, segmented by device brands, regions, and time periods. However, trends in device usage vary significantly across regions, and some devices are disproportionately underutilized despite high registration numbers.

Objective: Identify device brands used by PhonePe users.

Insights:

Query:1.Top Device Brands by Registered Users

- Brands like Xiaomi, Samsung, and Vivo have the highest registered users.
- These brands dominate the Indian smartphone market.



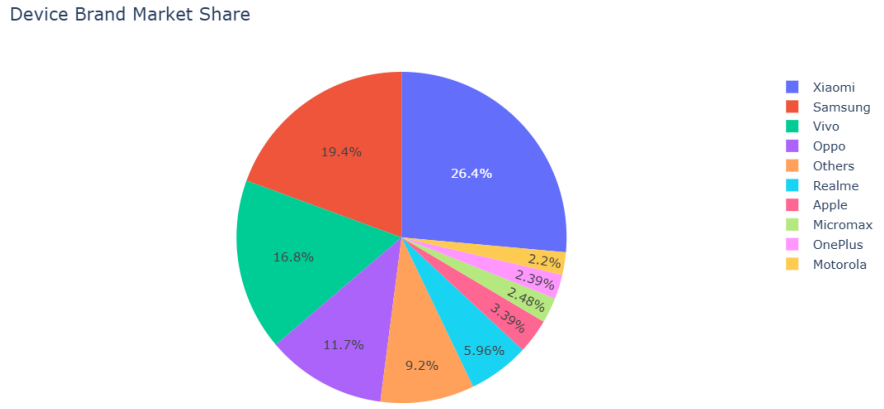
Key Insight:
Few brands dominate the user base, reflecting overall smartphone market trends.

Recommendation:
Optimize app performance for top devices and ensure compatibility across these brands.

Query:2. Device Brands by User Percentage

- A few brands account for the majority of users.

- This indicates device brand concentration in the user base.

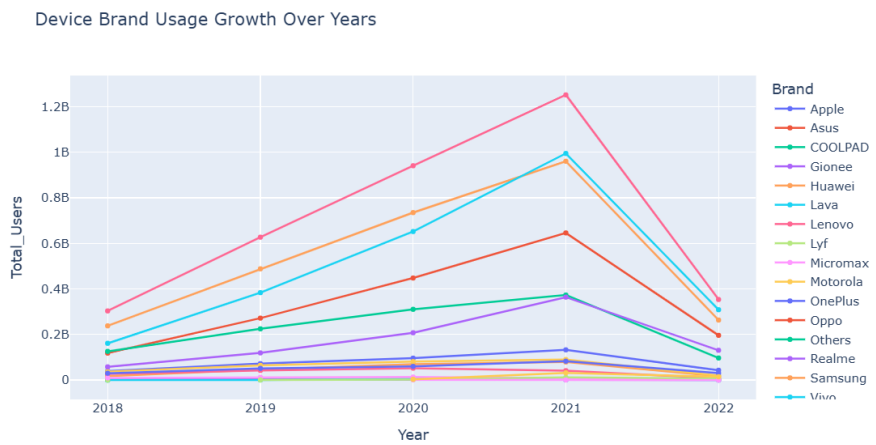


Key Insight:
User distribution is concentrated among limited brands, showing dependency on key manufacturers.

Recommendation:
Collaborate with major brands for promotions and pre-installed app integrations.

Query:3. Device Brand Usage Growth Over Year

- Registered users across major device brands increased steadily.
- Shows growth of smartphone usage and mobile payment adoption.

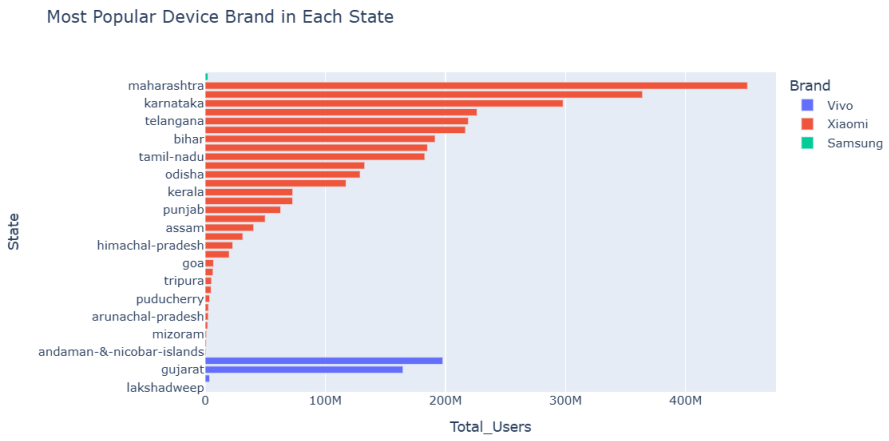


Key Insight:
Device usage is increasing steadily, indicating rising smartphone penetration.

Recommendation:
Continue mobile-first strategies and enhance app features for better user experience.

Query: 4. Top Device Brand in Each State

- Different states show preference for different smartphone brands.
- This helps PhonePe optimize app performance for popular devices.

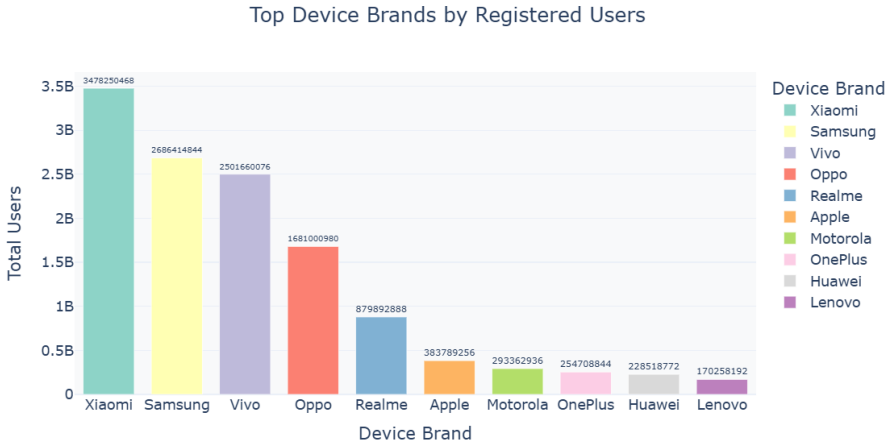


Key Insight:
Different states prefer different device brands, showing regional variation.

Recommendation:
Customize app optimization and marketing based on state-wise device preferences.

Query:5. Top 10 Device Brands Comparison

- A small group of brands dominates the PhonePe ecosystem.
- PhonePe can focus testing and optimization efforts on these brands.



Key Insight:
A small number of brands dominate overall usage.

Recommendation:
Focus testing, updates, and UI optimization on these major brands for better performance.

3. Insurance Penetration and Growth Potential Analysis

Scenario

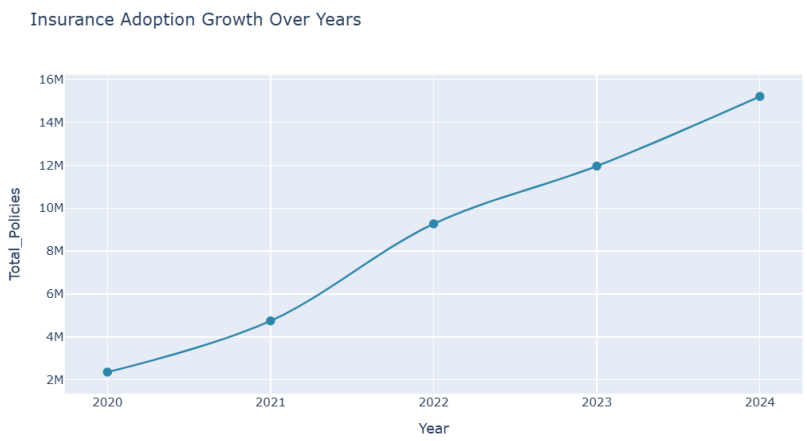
PhonePe has ventured into the insurance domain, providing users with options to secure various policies. With increasing transactions in this segment, the company seeks to analyze its growth trajectory and identify untapped opportunities for insurance adoption at the state level. This data will help prioritize regions for marketing efforts and partnerships with insurers.

Objective: Analyze growth of insurance transactions.

Insights:

Query: 1. Insurance Growth Over Years

- Insurance transactions have increased gradually.
- This indicates growing awareness of digital insurance services.

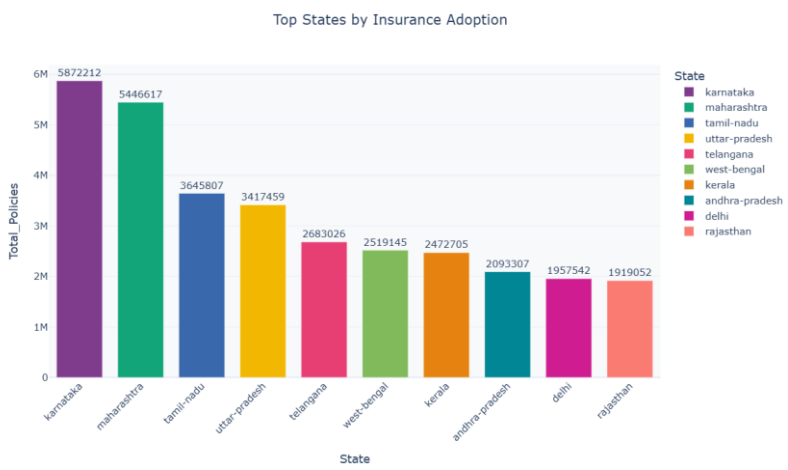


Key Insight:
Insurance adoption is increasing gradually over time.

Recommendation:
Increase awareness through campaigns and simplify the onboarding process.

Query:2. Top States by Insurance Policies

- Some states show higher policy adoption.
- These states represent strong insurance markets.



Key Insight:

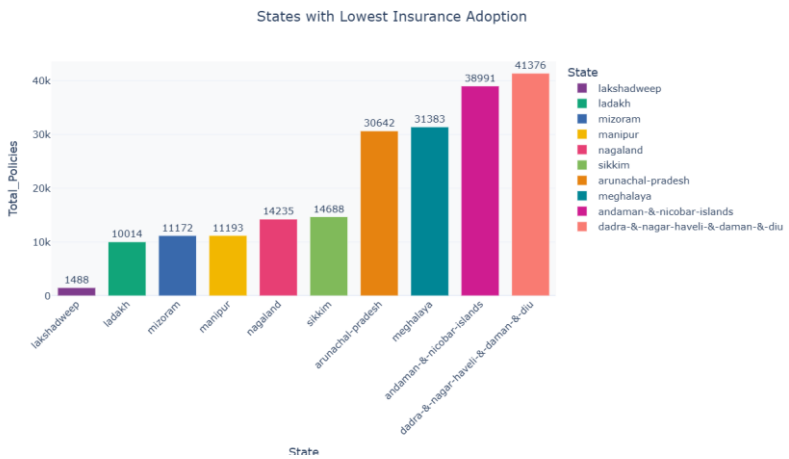
Certain states show strong insurance adoption.

Recommendation:

Strengthen partnerships with insurers in these high-performing regions

Query:3. Lowest Insurance Adoption States

- Several states have very low insurance adoption.
- These regions present growth opportunities for marketing campaigns.



Key Insight:

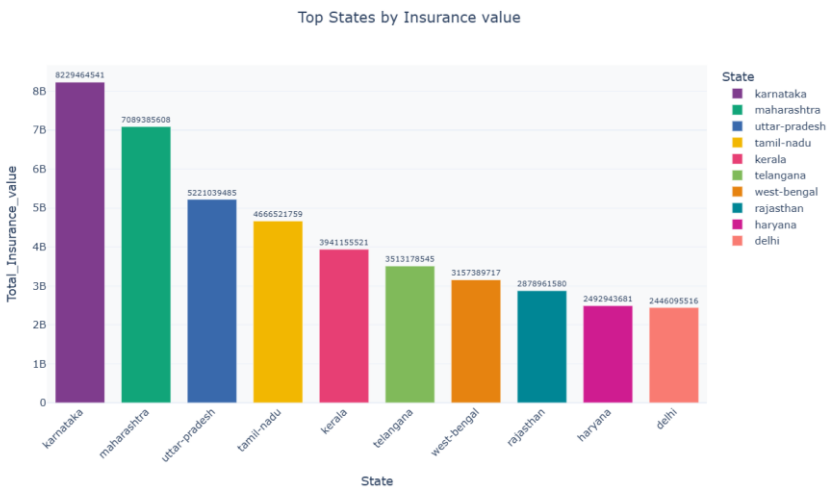
Some states show very low adoption, indicating untapped markets.

Recommendation:

Run targeted awareness campaigns and offer incentives in these regions.

Query:4. Top States by Insurance Value

- Certain states contribute significantly to total insurance value.
- Indicates higher-value policies purchased in these regions.



Key Insight:

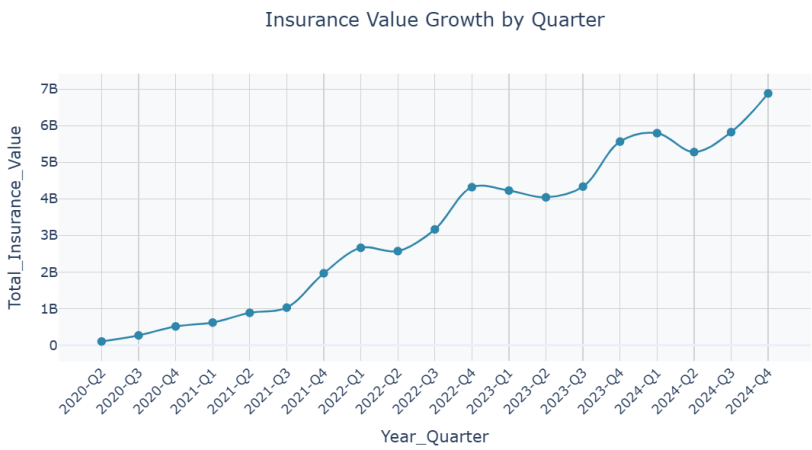
Some states contribute higher insurance value, indicating premium policy purchases.

Recommendation:

Promote premium insurance products in these regions.

Query:5. Insurance Adoption Trend by Quarter

- Insurance purchases fluctuate across quarters.
- This suggests seasonal purchasing patterns.



Key Insight:

Insurance purchases vary across quarters, indicating seasonal trends.

Recommendation:

Launch campaigns during peak periods to maximize conversions.

4. Transaction Analysis for Market Expansion

Scenario

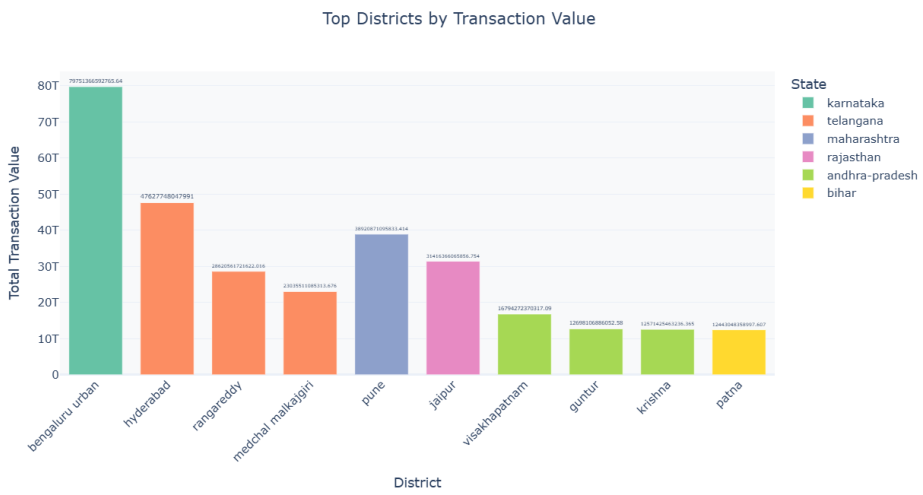
PhonePe operates in a highly competitive market, and understanding transaction dynamics at the state level is crucial for strategic decision-making. With a growing number of transactions across different regions, the company seeks to analyze its transaction data to identify trends, opportunities, and potential areas for expansion.

Objective: Identify districts with high and low transaction performance.

Insights:

Query:1. Top Districts by Transaction Value

- A few districts contribute significantly to total transaction value.
- These regions represent high engagement areas.



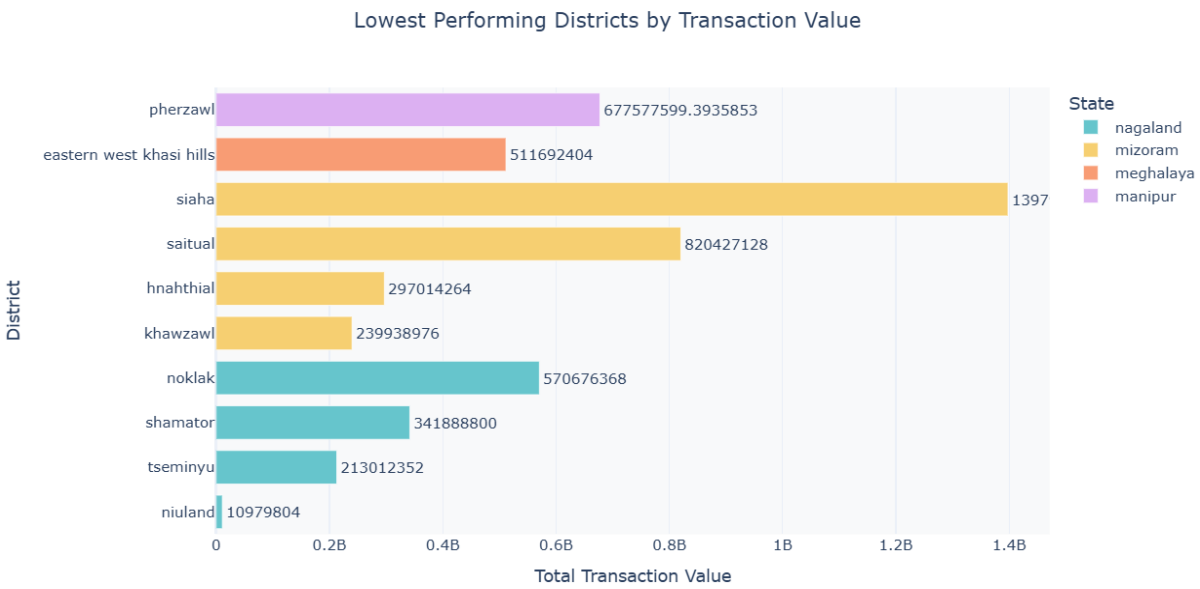
Key Insight:
Few districts contribute significantly to total transactions.

Recommendation:
Focus business expansion and partnerships in these high-performing districts.

Query:2. Lowest Performing Districts

- Some districts show very low transaction activity.

- These districts may lack digital awareness or infrastructure.



Key Insight:

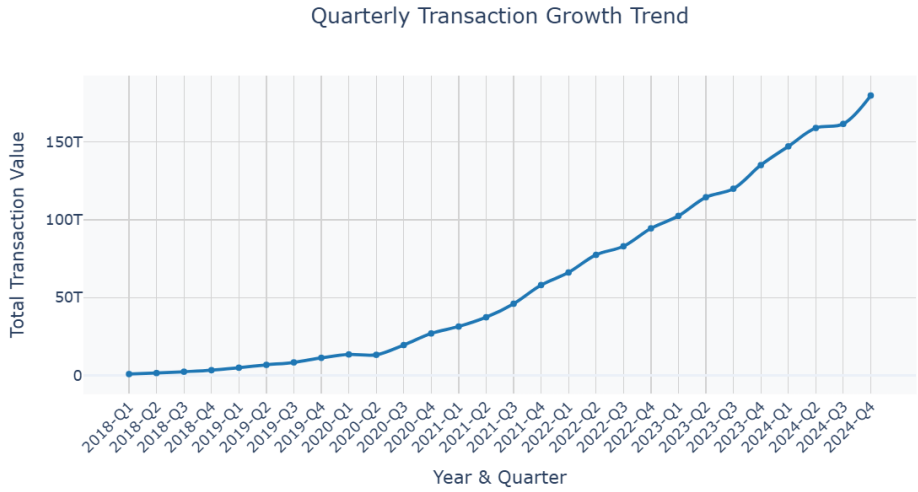
Some districts have low transaction activity due to limited awareness.

Recommendation:

Improve digital literacy and infrastructure awareness in these regions.

Query:3. Quarterly Transaction Growth Trend

- Transactions increase during certain quarters.
- This indicates seasonal usage patterns.



Key Insight:

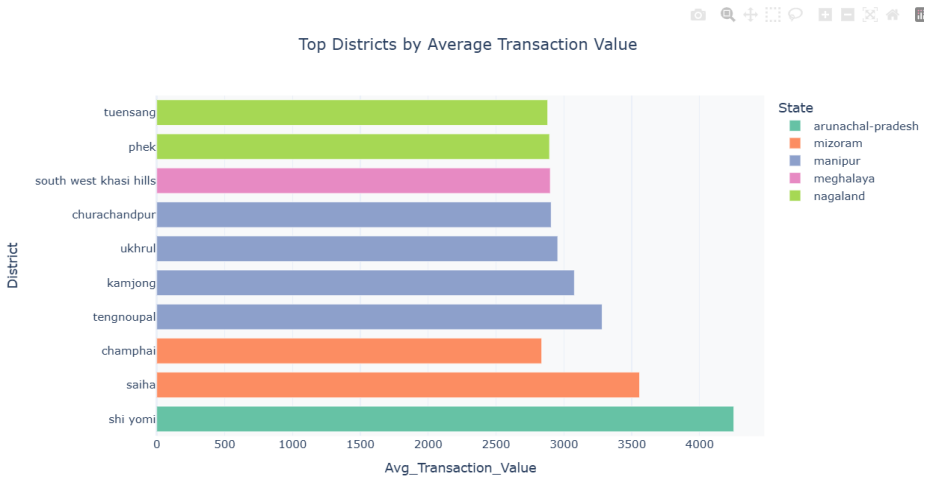
Transactions fluctuate across quarters, showing seasonal patterns.

Recommendation:

Plan marketing campaigns during high-demand periods.

Query:4. Average Transaction Value per District

- Some districts have higher average transaction values.
- These areas may represent higher purchasing power.



Key Insight:

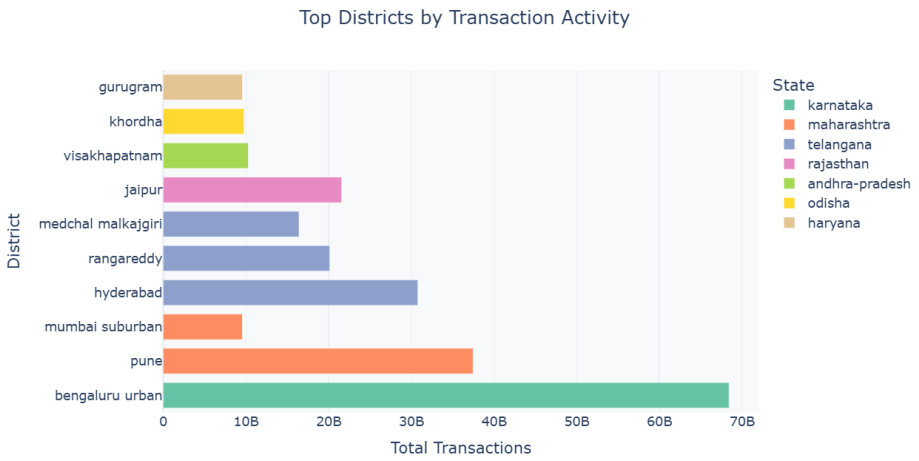
Some districts show higher average values indicating higher spending capacity.

Recommendation:

Target premium services and high-value offers in these districts.

Query:5. High Transaction Count but Lower Value Districts

- Some districts have many transactions but smaller values.
- These regions may reflect frequent small payments such as utility bills or recharge



- **Key Insight:**
  Some regions have frequent low-value transactions. Small districts often show higher averages because, They may have low transaction count, But some transactions are high value. So the average becomes large.
- This insight actually means:These districts have high-value transactions, But low volume
  Recommendation:
  Introduce micro-payment offers and cashback for small transactions. PhonePe could target high-income users or merchants there.

5. User Engagement and Growth Strategy

Scenario:

PhonePe seeks to enhance its market position by analyzing user engagement across different states and districts. With a significant number of registered users and app opens,

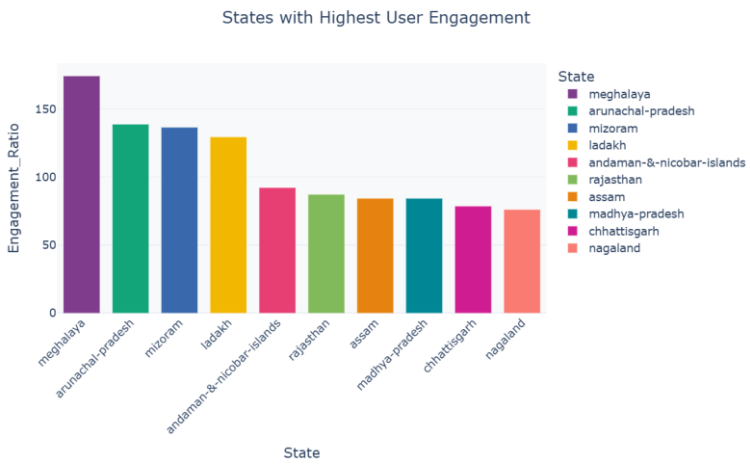
understanding user behaviour can provide valuable insights for strategic decision-making and growth opportunities.

Objective: Analyze user engagement based on registrations and app usage.

Insights:

Query:1. States with Highest User Engagement

- Some states show a higher ratio of app opens compared to registered users.
- This indicates active and loyal users.



Key Insight:

Some states show high app usage relative to user base.

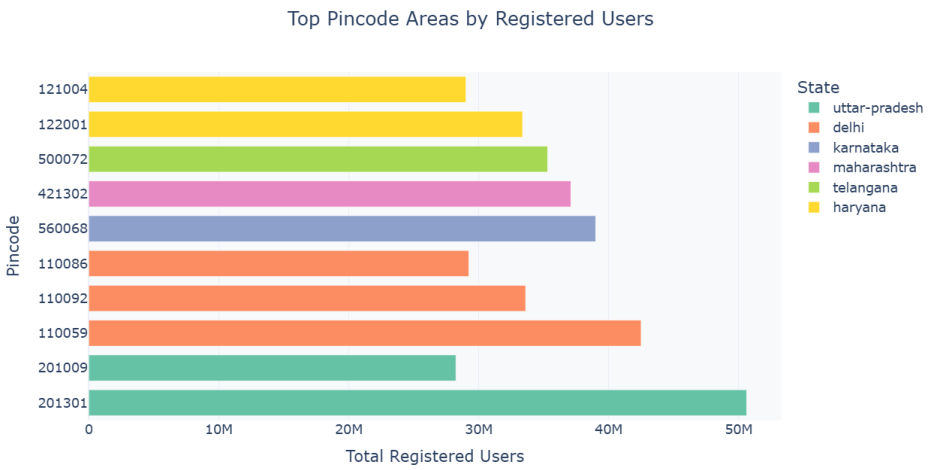
Recommendation:

Leverage these regions to launch new features and beta testing.

[Small states like: Meghalaya, Mizoram, Arunachal Pradesh may have fewer users, but those users open the app very frequently. So their engagement ratio becomes high.]

Query:2. Pincode Areas with Highest User Concentration

- Certain pin codes have a high number of registered users.
- These areas represent dense user markets.



Key Insight:

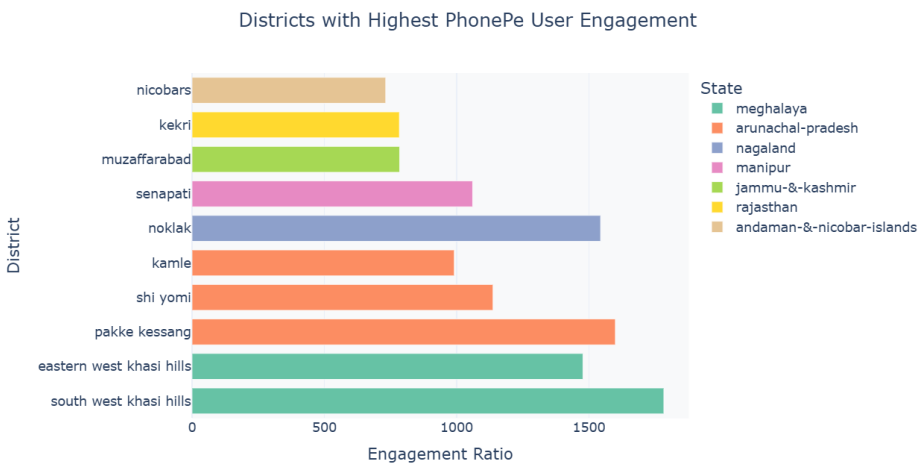
Certain pincodes have dense user concentration.

Recommendation:

Target these areas for localized promotions and merchant onboarding.

Query:3. Districts with Highest User Engagement

- Some districts show strong engagement levels.
- These areas are ideal for launching new features or campaigns.



Key Insight:

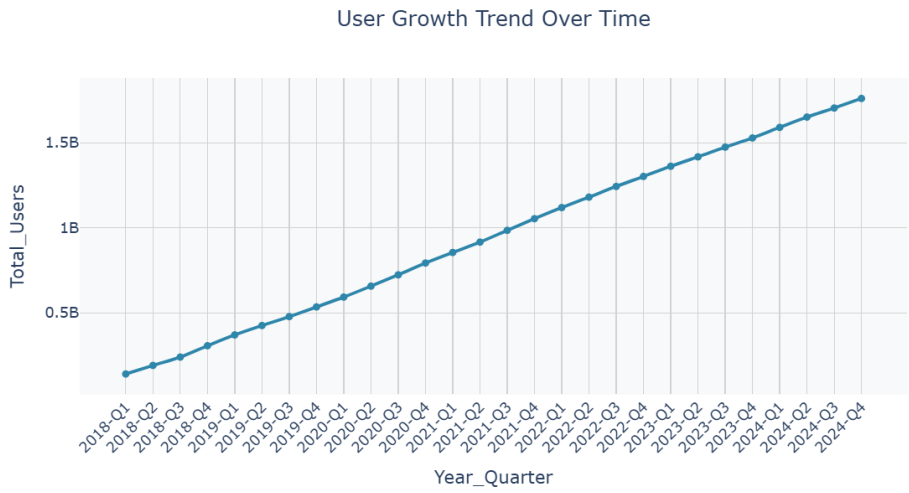
Some districts show strong user activity and engagement.

Recommendation:

Focus campaigns and partnerships in these districts.

Query:4. User Growth Trend by Year and Quarter

- Registered users have grown consistently across years.
- Shows increasing adoption of PhonePe services.



Key Insight:

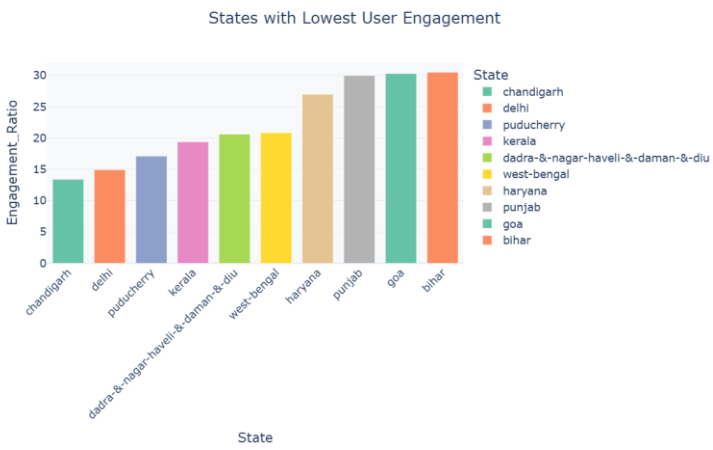
User base is steadily increasing over time.

Recommendation:

Continue user acquisition strategies and referral programs.

Query:5. States with Low Engagement

- Some states have low engagement despite registered users.
- PhonePe can improve engagement through targeted promotions and awareness campaigns.



Key Insight:

Some states have low engagement despite registrations.

Recommendation:

Improve engagement through offers, notifications, and awareness campaigns.

Dashboard Development:

An interactive Streamlit dashboard was created to display the analysis results. The dashboard allows users to explore:

- Transaction trends
- Device brand usage
- Insurance growth
- State-wise performance
- User engagement metrics

The dashboard makes the data interactive and easier to interpret.